RCOEM
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ML BASED PROSTHETIC HAND USING EMG SENSOR

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SUMMARY

INTRODUCTION



A prosthesis is an artificial body part that replaces some of the function and appearance of the missing part.

Highlights:

- Motion Replication: EMG sensors enable muscle-based gesture control.
- Machine Learning Integration: CNN model classifies hand movements and finds the nearest match if a random input is detected.
- User Empowerment: Provides a functional, responsive aid for limb-affected individuals.

BACKGROUND OF THE STUDY

Existing Prosthetic Arms Issues

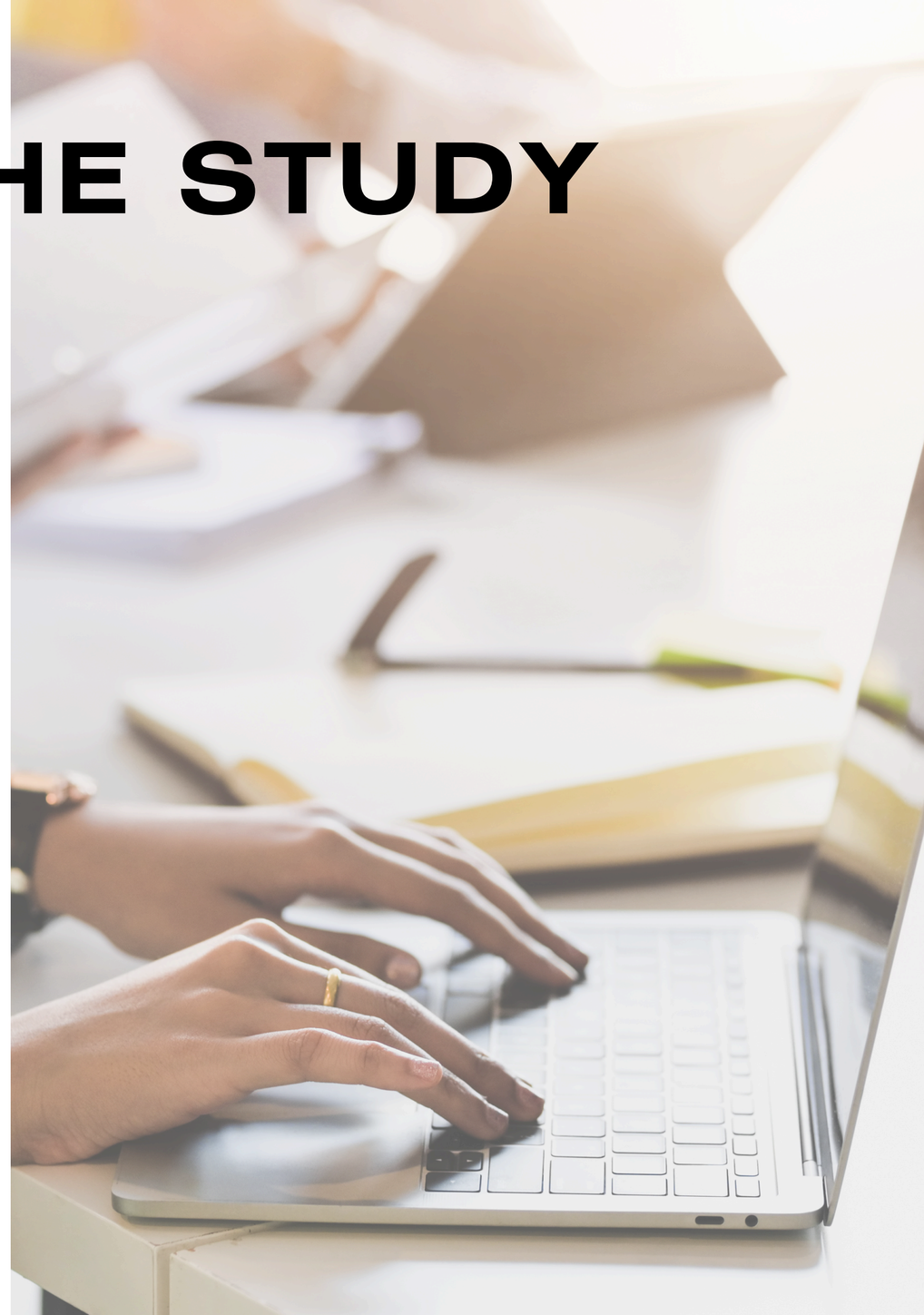
- Highly Expensive i.e. costing upto 4-7 lakhs
- Weight and comfort

Critical Needs

- Cost Effective
- Custom-made

Possible Solutions

- Myoelectric Prosthetic Arm
- 3D-Printed Prosthetic Arm



EXISTING MODELS

Company	Model Name	Cost (approx.)	Weight (g)	Limitation
Össur (Touch Bionics)	i-limb Quantum	$50,000 - $60,000	599	Requires frequent charging, expensive maintenance.
Ottobock	Michelangelo Hand	$70,000 - $80,000	510	Limited precision for fine motor tasks.
Vincent Systems	Vincent Hand	$50,000 - $70,000	410	Lower force capacity, limited availability.
Open Bionics	Hero Arm	$3,000 - $5,000	1,100	Limited grip strength, customizable but fewer actuation options.
DEKA Research	LUKE Arm	$100,000+	1,800	Heavy compared to other models, high cost.

OBJECTIVES

To design Electro-Mechanical System controlled through Musco-skeletal system.

01

NON-INVASIVE

02

BATTERY OPERATED

03

LIGHT WEIGHT

04

DURABLE AND FLEXIBLE

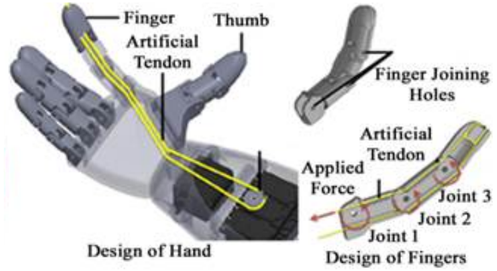
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COMPACT DESIGN

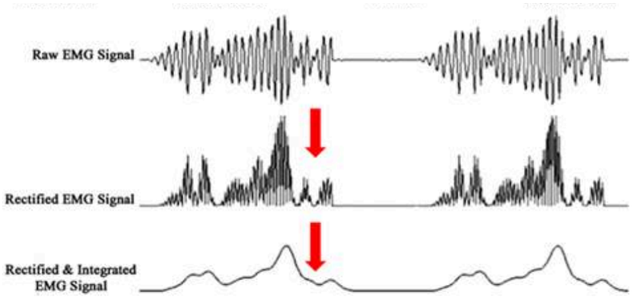
METHODOLOGY



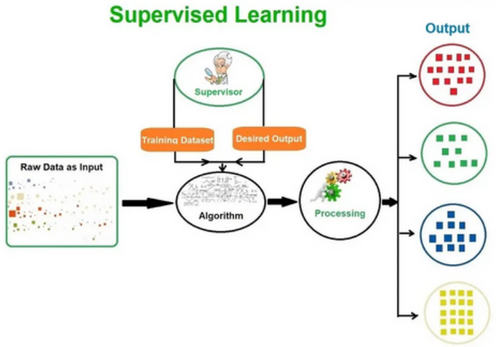
3D PRINTING OF PROSTHETIC ARM



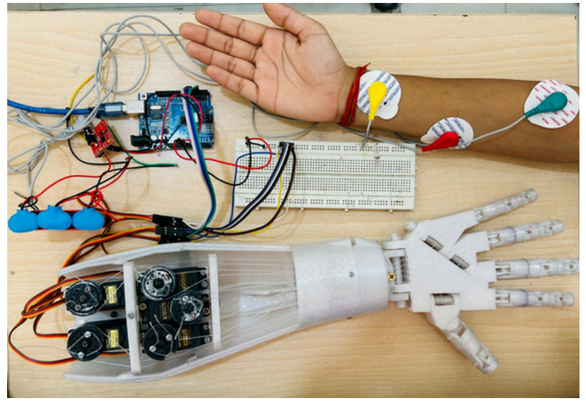
WORKING OF PROSTHETIC



ANALYSIS



IMPLEMENTATION OF MACHINE LEARNING

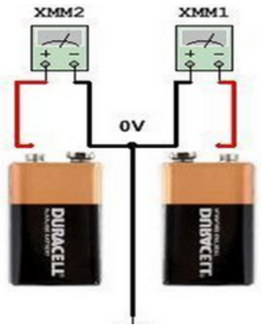


PROTOTYPE DEVELOPMENT

COMPONENTS



EMG MUSCLE SENSOR V3



POWER SUPPLY



ARDUINO UNO



SERVO MOTOR

COMPONENTS ASSEMBLED:-

- EMG muscle sensor v3
- Arduino Uno
- Servo motors (MG995R) (5)
- Power supply (± 9V)

COMPONENTS SPECIFICATIONS

	Emg muscle sensor v3	Arduino uno	servo motors (MG995r)	Power Supply
Current Range	1-15 mA	10mA - 1A	10 - 190 mA	12 Ah
Voltage	± 9 volts	6 volts	5 volts	9 volts
Weight(g)	8 grams	25 grams	55 grams	190 grams

COMPONENT SPECIFICATIONS

EMG MUSCLE SENSOR V3

- Bandwidth :- 20- 500 Hz
- Input Voltage :- 2-5 mV
- Built-in analog processing circuitry for noise reduction and filtering.

ARDUINO UNO

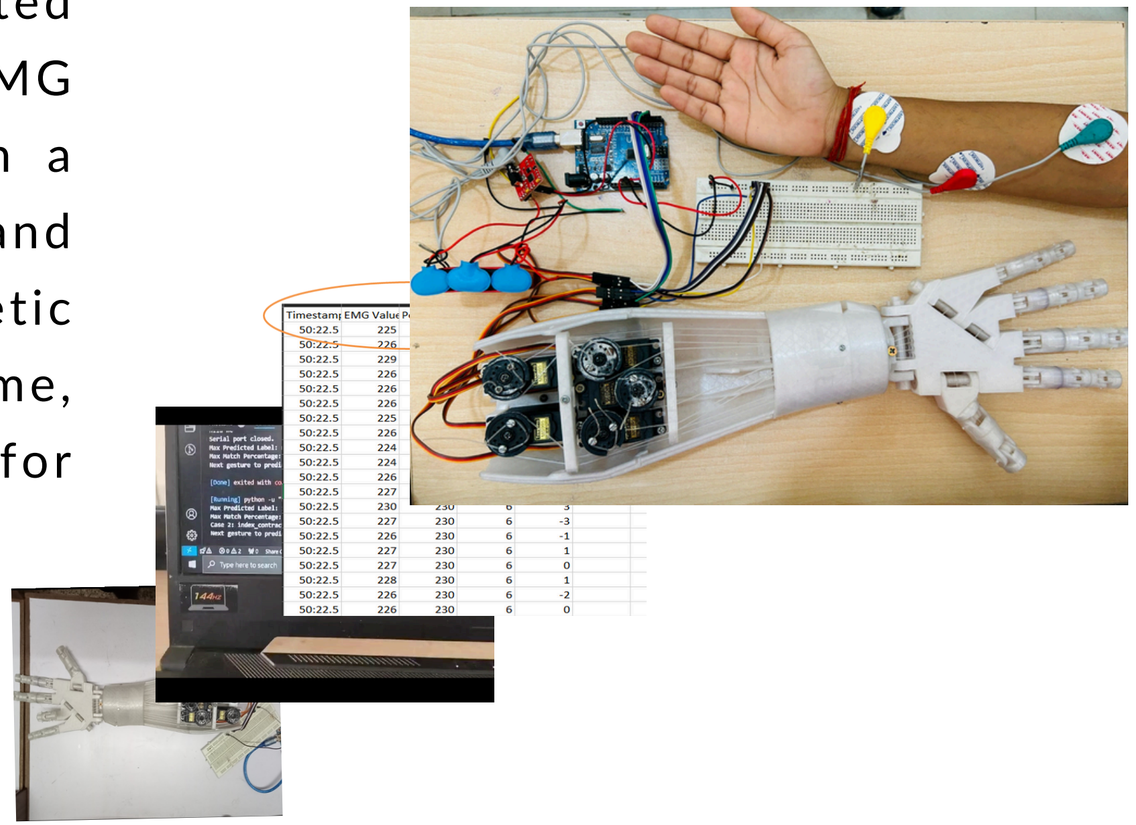
- Clk speed :- 16 Mhz
- Default clk speed :- 4Mhz

SERVO MOTORS (MG995R)

- Torque : 8.5 kgf.cm at 5v
- Total weight :- 5 X 55 grams

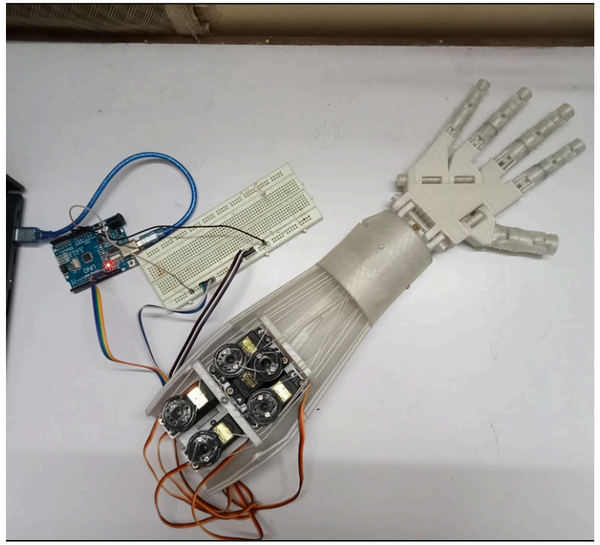
WORK DONE

This project involves designing a 3D-printed prosthetic hand controlled by a 3-electrode EMG sensor. Custom datasets were created to train a machine learning model to detect specific hand gestures from EMG signals, allowing the prosthetic to mimic natural hand movements in real-time, providing an affordable and responsive solution for limb-affected individuals.

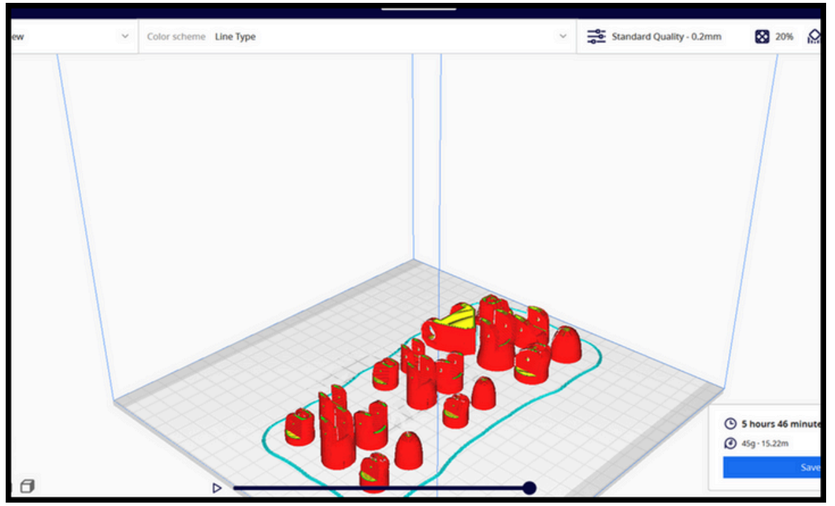


3D PRINTED PROSTHETIC HAND

- Designing and assembling of design in Cura
- 3D printing of Prosthetic hand with PLA filament
- Assembling of Servo Motors
- Creating connections through strings for movement of fingers



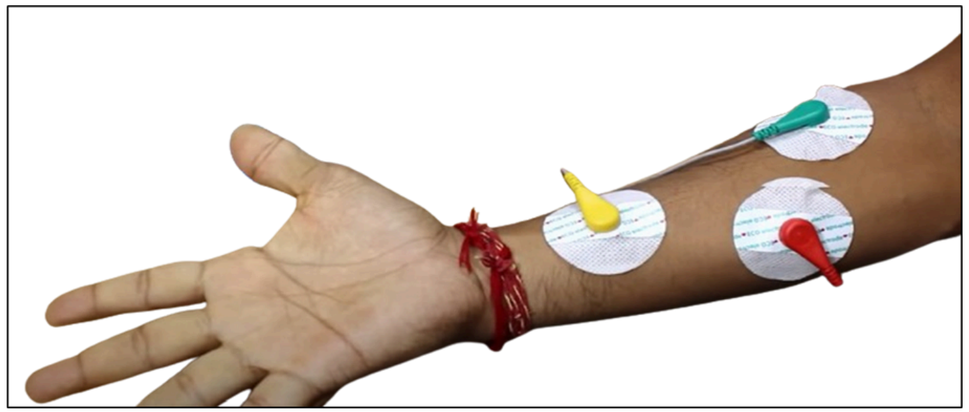
3D PRINTING OF PROSTHETIC ARM



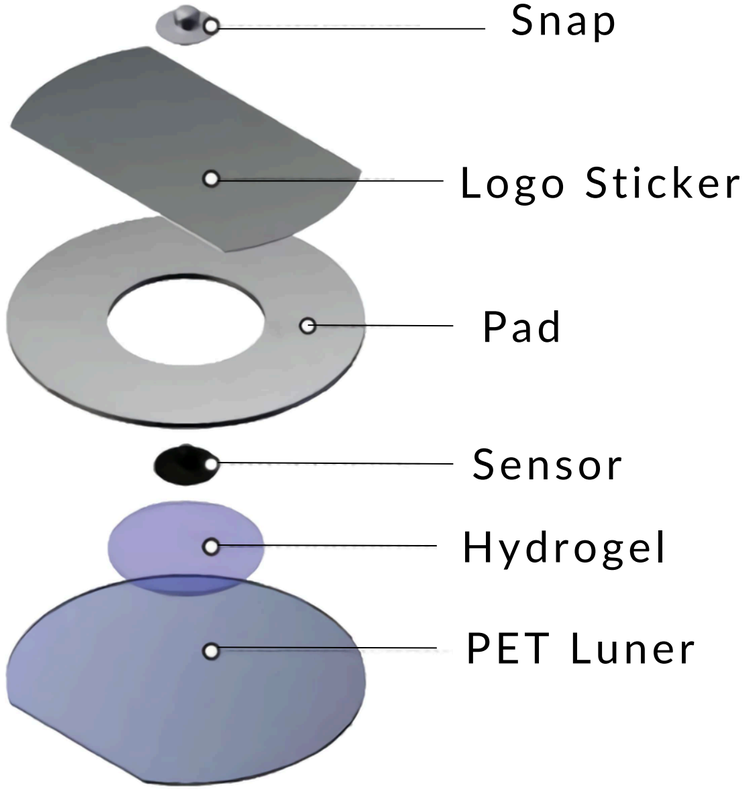
CURA SOFTWARE

PLACEMENT OF EMG SENSOR V3

- Literature research
- Study of EMG leads
  - The hydrogel is the only conductive component o an electrode that contacts the skin.
  - Most currently used hydrogels are liquid hydrogels, i.e., soft gel scaffolds filled with wate and salt



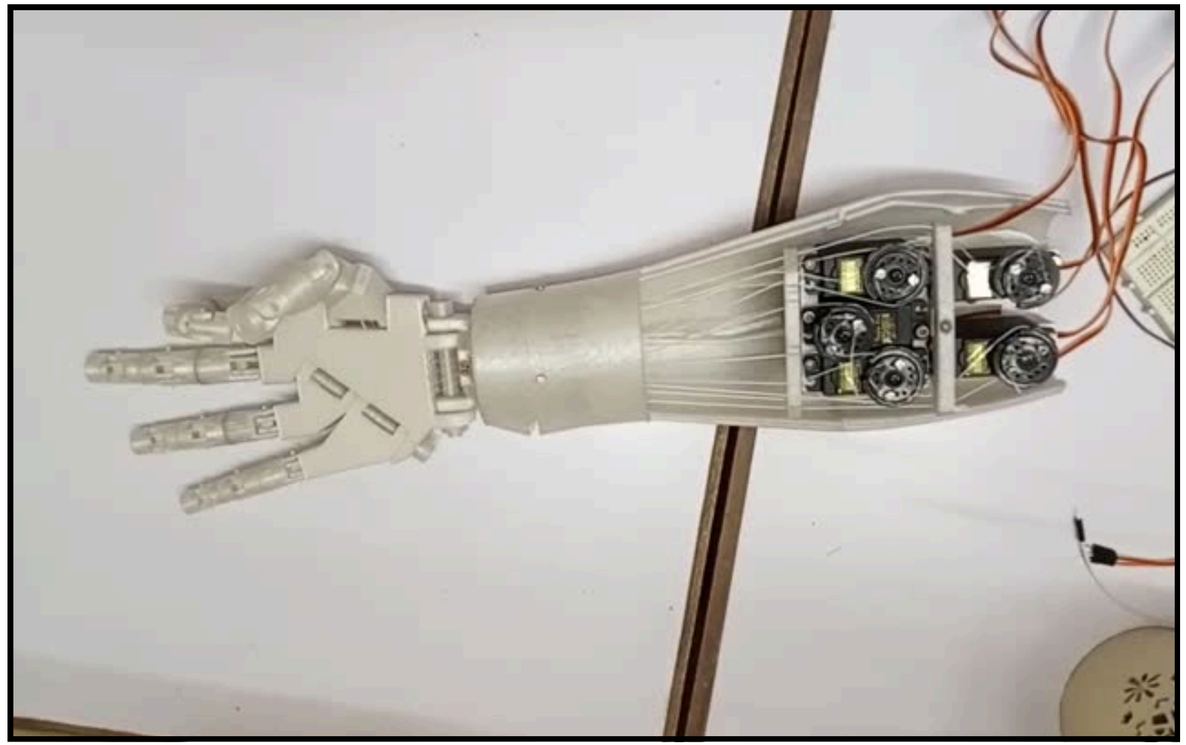
PLACEMENT OF EMG SENSOR



EMG LEAD

ISOLATED MOVEMENT OF PROTOTYPE

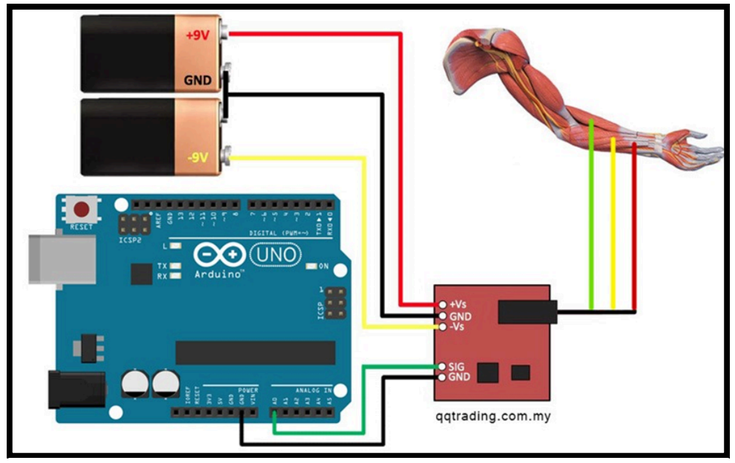
- Assembling of Servo motors for each finger
- Attaching String for accurate movement
- Interfacing Servo motor with Arduino Uno
- Code for Servo motor for each gesture



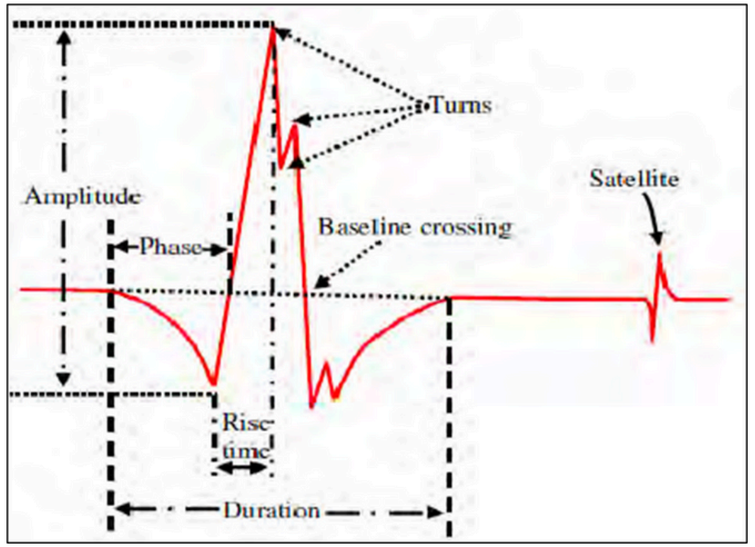
MECHANICAL WORKING OF PROSTHETIC HAND

INTERFACING OF EMG

- Circuit Formation
- Adjusting Power Supply module for EMG Data Collection
- Getting Accurate EMG Signal
- Monitoring of EMG Signal



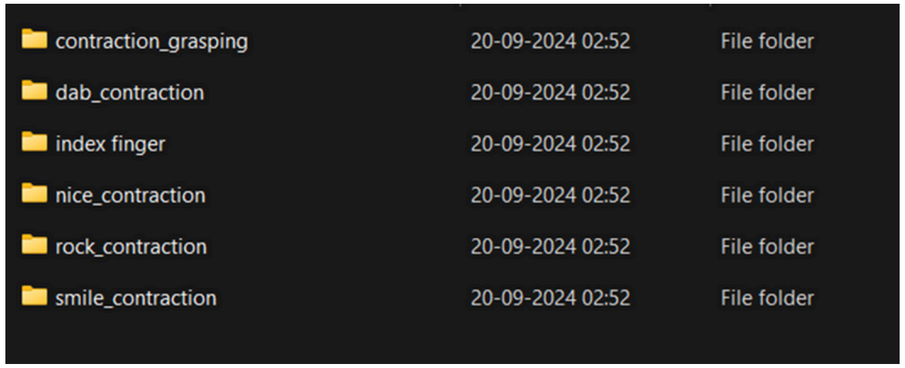
EMG INTERFACED WITH ARDUINO UNO



IDEAL EMG SIGNAL

BUILDING A COMPREHENSIVE DATASET

- EMG value along with other parameters stored in .csv file.
- Dataset for 6 different gestures are build.
- Frequency enhanced upto 500Hz.
- 5 datasets are prepared for each gesture performed for 4secs(constant for each gesture).
- Therefore, total of 10,000 data points(for each gesture) are used for training machine learning model.



DATASET FOR 6 GESTURES

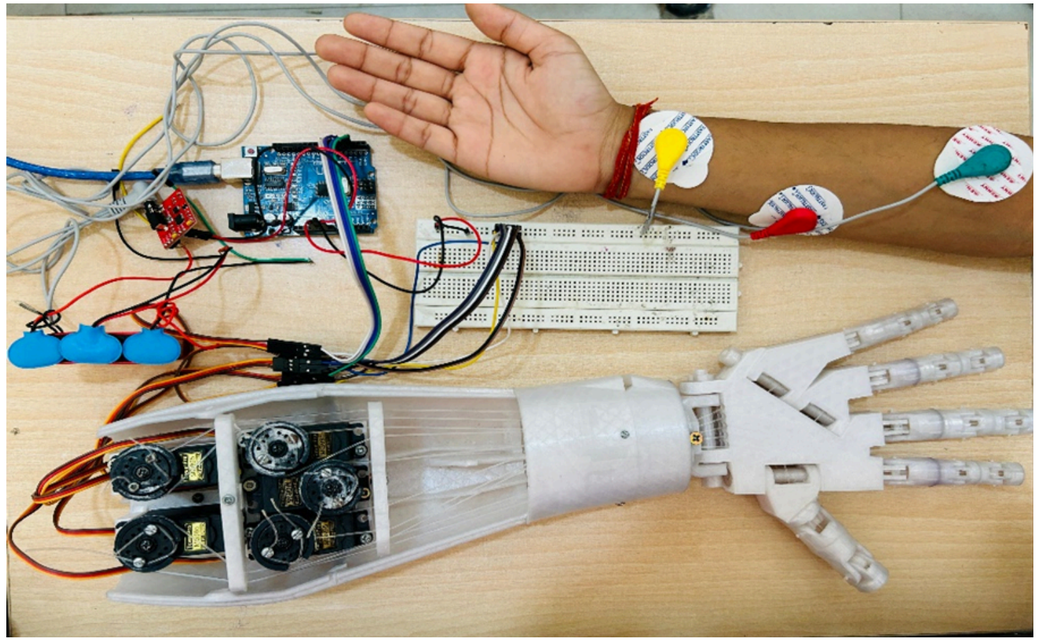
Timestamp	EMG Value	Peak Value	Amplitude	Phase
50:22.5	225	225	0	0
50:22.5	226	226	1	1
50:22.5	229	229	4	3
50:22.5	226	229	4	-3
50:22.5	226	229	4	0
50:22.5	226	229	4	0
50:22.5	225	229	4	-1
50:22.5	226	229	4	1
50:22.5	224	229	5	-2
50:22.5	224	229	5	0
50:22.5	226	229	5	2
50:22.5	227	229	5	1
50:22.5	230	230	6	3
50:22.5	227	230	6	-3
50:22.5	226	230	6	-1
50:22.5	227	230	6	1
50:22.5	227	230	6	0
50:22.5	228	230	6	1
50:22.5	226	230	6	-2
50:22.5	226	230	6	0
50:22.5	227	230	6	1
50:22.5	228	230	6	1
50:22.5	232	232	8	4
50:22.5	231	232	8	-1
50:22.5	229	232	8	-2
50:22.5	229	232	8	0
50:22.5	230	232	8	1

PARAMETERS/ DATA ACQUIRED

MACHINE LEARNING MODEL TRAINING (CNN)

- Dataset created are trained on CNN machine learning model.
- Different classes are formed for corresponding gesture.
- Epoch helps in training the model multiple times through the provided dataset
- Binary file (.joblib) file is formed stored all the trained data for classification.
- Testing of machine learning model is done by providing unknown data as input.
- The model finds the max-match of the data, and thus detects the corresponding class/gesture.

PROTOTYPE



PROTOTYPE

- Circuit design is designed for Interfacing both EMG sensor and servo motor (used for actuation of prosthetic hand).
- Code generated for combining machine learning model, EMG sensor data acquisition and servo motor actuation.
- Real-time data is acquired from EMG sensor, fed to model and thus gesture is detected.
- Gesture detected is fed back to controller for actuation of desired servo motor.
- 3 gestures have been defined accurately and performed by the prosthetic hand.

SUMMARY

Key Features

- Cost-effective
- Custom-Dataset Creation
- Gesture Recognition
- Real-time Control

Application

- Customized Dataset Creation
- Machine Learning for Gesture Detection
- Adaptive to User Variability

Novelty

- Rehabilitation Device
- Assistive Device for Disabled
- Human-Machine Interface

THANK YOU